Positional Faithfulness Theory and the Emergence of the Unmarked: The Case of Kagoshima Japanese

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Main Goal: (i) to analyze the consonantal reduction observed in Kagoshima Japanese in light of Positional Faithfulness Theory (Beckman 1998) and (ii) to critically assess the independent status of the Coda Condition (Ito 1986) in linguistic theorizing.

1. Introduction

1.1. Background

The Coda Condition

☞ Many natural languages disallow coda consonants which are not place-linked to a following onset consonant (see Ito 1986, 1989; Ito and Mester 1994 *inter alia*).

(1) A restriction on coda consonants

Well-Formed		Ill-Formed
happa	'a leaf'	hatpa
katta	'bought'	kapta
sakkaa	'soccer'	saktaa

☞ This ubiquitous restriction on coda consonants has been analyzed as the realization of "the Coda Condition" (since Ito 1986) that prohibits an independent place node linked to a coda consonant.

(2) Coda Condition

$$\begin{array}{c} *\mathrm{C}\,]\,\sigma \\ | \\ [\mathrm{PLACE}] \end{array}$$

Positional Faithfulness Theory

☞ Beckman (1998) obviates the Coda Condition *per se*, developing Positional Faithfulness Theory (PFT). PFT maintains that psychologically or semantically prominent positions are governed by "privileged" faithfulness constraints. These privileged positions include syllable onset (as opposed to syllable codas), word-initial syllables, stressed syllables, and roots (as opposed to affixes) and possibly others.

(3) Universal Positional Faithfulness Schema
Faith<onset> >> Faith

☞ The coda condition can be derived by sandwiching *PLACE by these two kinds of faithfulness constraints: Ident(place)<onset> >> *PLACE >> Ident(place).

(4) Deriving "Coda Condition Effect"

/kapta/	Ident(pla)<onset>	*PLACE	Ident(pla)
[kapta]		***!	
☞[katta]		**	*
[kappa]	*!	**	*

1.2. The goal

Our goal in this talk: The ranking above is necessary to explain the inventory restriction of coda consonants in Japanese. Given this ranking, what would happen if a coda consonant must stand alone (i.e., hence cannot be place-linked)? It is predicted that *PLACE >> Ident(place) necessitates that place specification is systematically lost. We argue that this prediction is in fact borne out. The empirical data is taken from Kagoshima Japanese.

1.3. Organization of the talk

Section 1: Introduction (already done)
Section 2: Data Presentation
Section 3: Optimality-Theoretic Analysis

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2. Data

2.1. Restriction on coda consonants

☞ Kagoshima Japanese, as in Tokyo Japanese, instantiates the coda condition effect. That is, coda consonants must share the place specification with the following onset sound.

(5) a. natta ‘became’
attʃi ‘there’
nanna ‘tear’
ʃinzo ‘heart’

b. kiʔne ‘fox’
suʔnaka ‘little’
maʔnoʔ ‘pine tree’

2.2. Apocope

☞ Kagoshima Dialect exhibits active apocope (the deletion of a word-final vowel) of high vowels (Haraguchi 1984; Kaneko 1998; Kibe 2002). This only takes place in Yamato-Stratum vocabulary (Kibe p.c).

☞ As a result of apocope, consonants placed word-finally are reduced to unmarked sounds (to the extent possible, in the sense that will be fleshed out shortly).

(6) Stop, Affricates: b, ts, tʃ, dz, dʒ, k, g => [ʔ]

tobu	toʔ	‘fly’
kutsu	kuʔ	‘shoes’
kutʃi	kuʔ	‘mouth’
midzu	miʔ	‘water’
adʒi	aʔ	‘taste’
kaki	kaʔ	‘persimmon’
ojogu	ojoʔ	‘swim’

(7) Fricatives: s, z, ʃ, ʒ=> s, ʃ

kwaʃi	kwaʃ	‘sweets’
kwaʒi	kwaʃ	‘fire’
usu	us	‘a morter’
kazu	kas	‘number’

(8) Nasal: n, (ɲ,) m => [N]

taɲi	taN	'valley'
inu	iN	'dog'
kami	kaN	'paper'

(9) Rhotic r => [j]

mari	maj	'ball'
çiru	çij	'noon'

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3. Analysis

3.1. Deriving Coda Restrictions in general

Basic Schema

☞ *PLACE is decomposed into the "place markedness sub-hierarchy" (Prince and Smolensky 1993; Beckman 1998; Alderete et al 1999; Lombardi 1998ab, 2001 among others):

(10) *Lab(ial), *Dor(sal), >> * Cor(onal)

☞ Given that *Cor is ranked the lowest, coronal consonants are least marked amongst the three places of articulation. Even less marked option, however, is not to have a place specification at all i.e., to best satisfy (10) is to remain placeless or to be linked to a following onset sound, thereby having no place specification of its own.

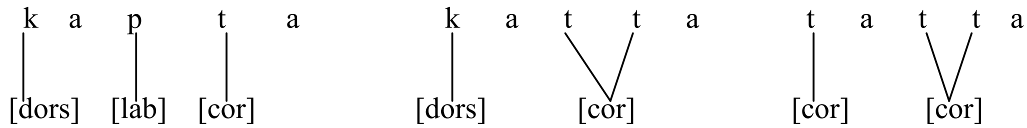
☞ The coda condition effect can be derived from the interaction of these two families of faithfulness constraints and the place markedness sub-hierarchy (10). The coda condition effect emerges when Ident(place)<onset> and Ident(place) together sandwich the place markedness sub-hierarchy. Taking a hypothetical input:

(11) The CodaCond Effect by PFT

/kapta/	Ident-Place<onset>	*LAB	*DOR	*COR	Ident-Place
a. kapta		*!	*	*	
b. ☞katta			*	*	*
c. tatta	*!			**	**

(12) Representation of each candidate

(a) (b) (c)



Placeless consonants

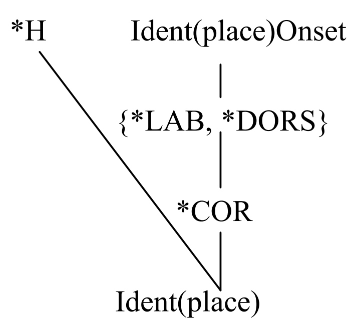
☞ In addition to place-linked consonants, Kagoshima Japanese allows a glottal stop in coda position. We assume that a glottal stop can appear as a coda by virtue of being placeless (see Clements 1985; Steriade 1987; McCarthy 1988; Clements and Hume 1995), and hence incurs no violation of any *PLACE-family constraint.

(13)

/kiʔne/	*LAB	*DOR	*COR	Ident(place)
a. ☞kiʔne		*	*	
b. kinne		*	*	*!

☞ What about [h], which is also considered to be placeless? This sound, however, is in fact prohibited in the Yamato-stratum except for word-initially (Kibe p.c). We hence assume that a constraint against [h] in this stratum is ranked so high that it allows the presence of [h] only in very limited environments. Following Beckman (1998; Chapter 2), word-initial syllables are assumed to be governed by positional-specific faithfulness constraints. This and only this faithfulness constraint dominates *H, the prohibition against [h].

(14) Interim Summary: Ranking deriving “ the Coda Condition” in Kagoshima Japanese



3.2. Apocope: the emergence of the unmarked

Apocope Taking Effect

☞ The impetus is arguably Final-C, which requires that every prosodic word end with a consonantal element. This constraint is satisfied insofar as a word ends in an obstruent, nasal, liquid or glide.

☞ It is only high vowels that delete word-finally. We adopt Pulleyblank's (1998ab) idea that faithfulness to some feature F depends crucially on F's sonority (see also Haraguchi 1984): the deletion (or any alternation) of more sonorous elements is more serious than that of less sonorous elements (cf. High Vowel Devoicing in Japanese).

(15)	Final-C:	A prosodic word must not end with a vocalic element (McCarthy and Prince 1994; McCarthy 1993, 2002a; Gafos 1998 and others)
	MaxNonHigh:	No deletion of non-high vowels (McCarthy and Prince 1995; Pullyblank 1998ab)
	MaxHigh:	No deletion of high vowels.

☞ These need to be ranked as MaxNonHigh >> Final-C >> MaxHigh.

(16) MaxNonHigh >> Final-C >> MaxHigh

/CVCV/ [-high]	MaxNonHigh	Final-C	MaxHigh
a. ☞CVCV		*	
b. CVC	*!		

/CVCV/ [+high]	MaxNonHigh	Final-C	MaxHigh
a. CVCV		*!	
b. ☞CVC			*

☞ In the following discussion, we ignore MaxNonHigh because the constraint is orthogonal to our discussion, and let Max represent MaxHigh.

Stops

☞ Given that the prohibition against a place specification is ranked higher than the faithfulness constraint regulating the identity preservation in coda positions, word-final consonants systematically lose their place when they are forced to stand alone word-finally, whether they be a labial, coronal or dorsal.

(17) Labial

/tobu/	Final-C	Max	Ident(pla) <onset>	*LAB	*DORS	*COR	Ident(pla)
a. tobu	*!			*		*	
b. tob		*		*!		*	
c. ☞toʔ		*				*	*
d. ʔoʔ		*	*!				**

(18) Coronal

/kutsu/	Final-C	Max	Ident(pla) <onset>	*LAB	*DORS	*COR	Ident(pla)
a. kutsu	*!				*	*	
b. kuts		*			*	*!	
c. ☞ kuʔ		*			*		*
d. ʔuʔ		*	*!				**

(19) Dorsal

/kaki/	Final-C	Max	Ident(pla) <onset>	*LAB	*DORS	*COR	Ident(pla)
a. kaki	*!				**		
b. kak		*			**!		
c. ☞kaʔ		*			*		*
d. ʔaʔ		*	*!				**

☞ This situation can be conceived of as an instance of *the emergence of the unmarked* (McCarthy and Prince 1994); by Final-C >> Max, generally prohibited sounds (i.e., word-final consonants) are forced to surface, in which case they must be least marked.

☞ Some considerations on other featural domains are necessary here: since a glottal stop is [-vcd], the change from voiced stops to [ʔ] incurs a violation of Ident(voi), and hence this constraint must be ranked below the place markedness sub-hierarchy

(20) *LAB, *DORS >> *COR >> Ident(voi)

/kadʒi/	*Lab	*Dors	*COR	Ident(voi)
a. kadʒ		*	*!	
b. ☞ kaʔ		*		*

Nasals

(21)

/kami/	Final-C	Max	Ident(pla) <onset>	*LAB	*DORS	*COR	Ident(pla)
a. kami	*!			*	*		
b. kam		*		*!	*		
c. ☞kaN		*			*		*

Fricatives

☞ Fricatives are not reduced to a placeless coda [h]. This has to do with the general ban against [h] in this dialect. Due to this prohibition against [h], as word-final fricatives, the most unmarked sounds (i.e., coronal fricatives) appear. We assume that [s] and [ʃ] are both coronal (Cairns 1988 and reference cited therein), and they are distinguished in terms of height: [s] is [-high] and [ʃ] is [+high].

(22)

/kwaʃi/	Final-C	Max	Ident(pla) <onset>	*H	*LAB	*DORS	*COR	Ident(pla)
a. kwaʃi	*!				*	*	*	
b. kwah		*		*!	*	*		*
c. ☞ kwaʃ		*			*	*	*	

(23)

/usu/	Final-C	Max	Ident(pla) <onset>	*H	*LAB	*DORS	*COR	Ident(pla)
a. usu	*!						*	
b. uh		*		*!				*
c. ☞us		*					*	

☞ Given the Richness of the Base (Prince and Smolensky 1993; Smolensky 1996; Kurisu 2000; McCarthy 2002b: 68-82) we need to make sure that any kind of fricatives can be a potential input, and have them all converge into coronal.

(24)

/uh/	Final-C	Max	Ident(pla) <onset>	*H	*LAB	*DORS	*COR	Ident(pla)
b. uh		*		*!				
c. ☞ us		*					*	*

☞ Indeed, any fricative sounds ([h], [ɸ], [ɕ], [ç] etc…) converge into coronal sounds in our ranking. Notice importantly that the coda condition alone on the other hand cannot account for this point. All fricatives with a place feature are equally bad w.r.t. the coda condition, and thus the coda condition is unable to differentiate the status of coronal fricatives from other fricatives. Therefore, dorsal and labial fricatives cannot become coronal fricatives due to the requirement of the coda condition.

☞ Coronal sounds are allowed to surface precisely because there is no less marked fricative i.e., coronals are most unmarked among other options.

☞ The real generalization that explains every aspect of coda consonants in this dialect is not that they are prohibited to have an independent place, as the coda condition requires. Coronal fricatives constitute crucial counterexamples. Given the *violability* of Optimality Theory (Prince and Smolensky; see also McCarthy and Prince 1993), this apparent counterexample to the coda condition might not constitute a crucial flaw of the coda condition approach; however, it still fails to explain (i) why only coronals are allowed for fricatives and (ii) why such an exceptional option is given only for fricatives.

☞ The approach we have developed readily provides an answer to these questions. Coronals are allowed for fricatives precisely because they are least marked options, and stops do not surface as coronals because there is a less marked option.

☞ To complete this discussion, the failure of neutralization between [s] and [ʃ] suggests that *HIGH is ranked lower than Ident(high).

(25) *HIGH: Feature specification in terms of [high] is prohibited.
Ident(high): The height specification in the input and output should be identical.

☞ It is also important to note here that fricatives do not become stops in order to avert a violation of *COR. Thus, Ident(cont) must dominate *COR. More generally, manner features are preserved (except for the case of a rhotic which becomes a glide; see below). This suggests that faithfulness constraints regulating manner features such as nasality are ranked relatively high i.e., higher than *COR at least.

(26)

/usu/	*LAB	*DORS	Ident(cont)	*COR
a. uʔ			*!	
b. ☞us				*

☞ Finally, coda devoicing is clearly observed in the case of word-final fricatives (e.g., /kazu/ => [kas]). The prohibition against voiced coda can also be accounted for by PFT by posing ranking below (see Beckman 1998: 28-51), where the ban against a voiced obstruent *VoiObs is flanked by two kinds of faithfulness constraints:

(27) Ident (voi)<onset> >> *VoiObs >> Ident(voi)

(28)

/do:kiN/	Ident (voi)<onset>	*VoiObs	Ident (voi)
a. ☞do:kiN		*	
b. to:kiN	*!		*

(29)

/kazu/	Ident (voi)<onset>	*VoiObs	Ident (voi)
a. kaz		*!	
b. ☞kas			*

Liquids
[r] placed word-finally becomes a non-syllabic high front glide [j]. We assume that this sounds forms a diphthong with the preceding nucleus vowel.

☞ There are no "placeless" liquids allowed in this dialect (in fact contrastive non-coronal liquids are extremely rare; Ladgeford and Madison 1996). Thus rather than becoming a placeless liquid, the language prefers to change the sound into a glide. This idea can be formally captured by the ranking: HavePlace(liquid) >> Ident(cons).

(30) HavePlace(liquid): A placeless liquid is prohibited.

☞ Why is [toj] more harmonic than [tor]? This is due to the ranking NoCoda >> Ident(cons). Note that a diphthong [Vj] satisfies Final-C while simultaneously avoiding the violation of NoCoda.

(31) NoCoda: a syllable must not have a coda.
:
(32) ([ℝ] represents a placeless liquid):

/tori/	HavePlace(liquid)	*COR	NoCoda	Ident (cons)
a. [toℝ]	*!			
b. [tor]		*	*!	
c. ☞ [toj]		*		*

☞ Concerning the ranking of NoCoda with respect to other constraints:
− it must be ranked below Final-C because the latter intrinsically necessitates a violation of NoCoda.
- the fact that this diphthongization strategy is not taken for stops and fricatives stems from Ident(son) >> NoCoda.
− the same pattern applies for nasals i.e., Ident(nas) >> NoCoda.

(33)

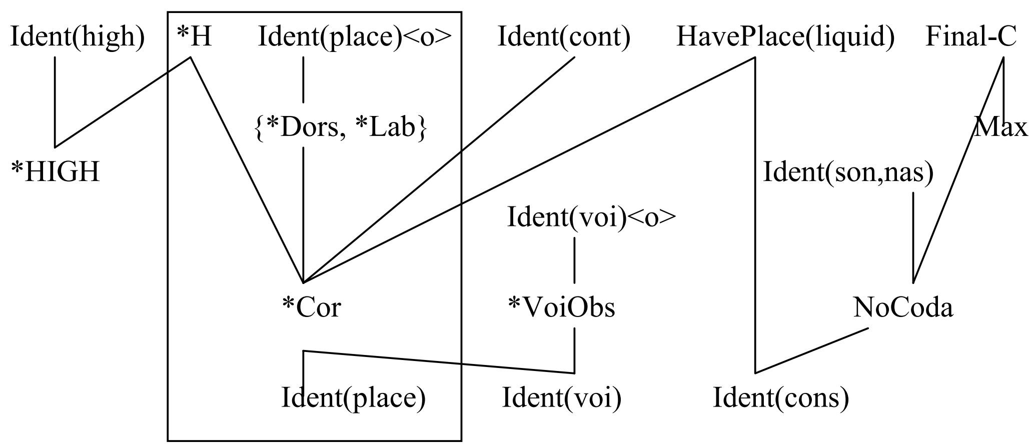
/kaki/	Ident (son)	NoCoda
a. ☞ kaʔ		*
b. kaj	*!	

(34)

/kami/	Ident (nas)	NoCoda
a. ☞ kaN		*
b. kaj	*!	

4. Summary

(35) Final Ranking



☞ The ranking in the box above is what we have motivated to account for the Coda Condition in Positional Faithfulness Theory.

- This part is not only compatible but constitutes the core to account for the reduction of word-final consonants as well.
- What the above ranking differs from the Tokyo dialect is that of Final-C and Max. Notice that the boxed part is also necessary for the Tokyo dialect.

Review of Main Points:

1. We have provided an empirical endorsement to PFT by showing that the consonant reduction in Kagoshima dialects can be elucidated by the independently motivated ranking within PFT.
2. The coda condition *per se is* eliminable from phonological theorizing.

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